Saurabh Nigam

Engineering Manager, Data & ML Platforms

Leads the platform teams behind Zeta's CDC replication and ML infrastructure, and the architecture of its lakehouse.

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Summary

Engineering Manager since 2024, with 10+ years in engineering across distributed systems and data platforms, 5 of them in technical leadership. Led Team Pontus, an 18-person organisation run through 2 team leads, setting its engineering process and owning architecture for bank-grade data platforms serving 5 to 6 enterprise banking customers. Rated Outstanding in two consecutive cycles (2024-25 and 2025-26). Now leads Zeta's Metis change data capture (CDC) replication and Elara ML infrastructure teams, drives architecture for an in-house lakehouse and builds AI agents and agent harnesses hands-on.

18 person org, led through 2 team leads · 2 consecutive Outstanding ratings, 2024-25 and 2025-26 ·
6 internal product lines on one Kubernetes operator · ~80% root-cause accuracy on evals, AI lineage agent

Experience

Engineering Manager, Data & ML Platforms · **Zeta**, Bengaluru, India — Jan 2024 to Present

Scope moved in 2026 from credit and loans data platforms (Team Pontus) to data replication and ML infrastructure platforms (Metis, Elara). Hands-on in design, architecture and code review (Java, Python).

Metis and Elara platforms 2026 to Present

- Leads two infrastructure platform teams: Metis for streaming change data capture (CDC) replication on Debezium, Kafka and Kubernetes operators; Elara for a Jupyter notebook platform and full-stack ML inference apps.
- Drives the architecture for an in-house lakehouse and transformation platform, with Spark and Redshift as interchangeable execution engines behind one interface.
- Defines CI/CD as a first-class product for internal teams, kept deliberately separate from application CI/CD.
- Owns roadmaps, SLAs and 24x7 reliability for both platforms with Product, SRE and Release Engineering.
- Heads end-to-end far-site disaster recovery for the Metis replication platform, which feeds downstream data and ML systems.

Team Pontus Jan 2024 to 2026

- Led an 18-person backend, platform and BI engineering org through 2 team leads, owning hiring, calibration and performance management, with 0% attrition and zero major production incidents in H1 2026.
- Set the engineering process across teams, from quarterly capacity planning and CI coverage gates to enforced data contracts that reduced schema breaks across hundreds of downstream services.
- Conceived and architected a declarative Kubernetes operator, built by the team, that replaced per-team operators, cut tenant onboarding from weeks to minutes and was adopted by 6 internal product lines.
- Directed the team that cut the largest aggregate models from 15+ to under 2 minutes and a large private bank's end-of-day batch from ~1.5 hours to ~35 minutes.
- Wrote the Redshift Serverless risk analysis that paused a premature migration, then guided a tenant-isolated rollout; the team halved base capacity from 128 to 64 RPUs, cutting infrastructure cost.

AI-native engineering 2024 to Present

- Built and open-sourced dbt-coverage with AI coding: a dbt and SQL test-coverage tool adopted at Zeta as a CI/CD quality gate.
- Launched a prompt-to-PR dbt harness for analytics engineers that turns a requirement into a tested, documented, quality-gated pull request, with an eval loop scoring agent skills against task suites.
- Built an AI lineage and incident-triage agent (LangChain, Claude) that uses dbt manifests to isolate root causes of pipeline failures, with ~80% accuracy on evals.
- Created a demand-deposit datamart for a new strategic client in one week with AI-assisted development, then anchored it technically until the client's dedicated team was hired.

Lead Software Developer · **Zeta** 2021 to Dec 2023

- Led and mentored a 5-engineer team that delivered a fault-tolerant loan servicing platform for HDFC Bank, one of India's largest private banks.
- Architected a high-throughput, event-driven financial transaction system on Kafka, Spring Boot, PostgreSQL and Redis.
- Improved SQL performance by ~50% through indexing, query refactoring and data modelling.
- Partnered with Product, SRE and Release Engineering on testing, monitoring and release processes for regulated banking systems.

Senior Software Engineer · **FarEye** 2016 to 2021

- Designed and built core Java microservices for real-time, global-scale logistics and delivery operations.
- Led the migration to Kubernetes, enabling containerised deployments and faster release velocity.
- Built a real-time Spark and Kafka data pipeline that improved system throughput by ~60%.
- Introduced a platform-wide alerting and notification system that improved observability and incident response.

Skills

ENGINEERING LEADERSHIP:	Manager of managers · Hiring and team building · Developing team leads · Org design · Performance management and rubric-driven calibration · Roadmap and capacity planning · Engineering process and SDLC · Architecture review · Stakeholder and executive communication · Cross-functional delivery
DATA PLATFORM:	dbt · Amazon Redshift (provisioned and Serverless) · Apache Spark · Trino (Presto) · Apache Airflow · Apache Iceberg · Delta Lake · AWS Glue and Amazon Athena · Lakehouse architecture · Data mesh and data contracts
PLATFORM ENGINEERING AND DISTRIBUTED SYSTEMS:	Apache Kafka · Debezium · Change data capture (CDC) · Kubernetes and Kubernetes operators · AWS · Infrastructure as code · CI/CD as a product (Jenkins, GitHub Actions) · Canary rollouts and quality gates · Disaster recovery · Observability, monitoring and SLAs
AI AND ML ENGINEERING:	Claude Code · MCP (Model Context Protocol) · Pydantic AI · LangChain · AI agents · Agent harness design · LLM evals and eval harnesses · Eval-driven engineering · Jupyter · ML inference apps
LANGUAGES AND BACKEND:	Java · Python · SQL · Spring Boot · REST APIs · PostgreSQL · Redis · Event-driven microservices · Docker
DOMAIN:	Regulated banking · Credit and loans · Payments · BIAN banking architecture · Multi-tenant fintech SaaS

Open Source

dbt-coverage, Author: Test-coverage and quality-gate tool for dbt and SQL projects, built end to end with AI coding and adopted at Zeta as a CI/CD gate.

llm-evals-engineering-course, Author: A 17-module, ~48-hour course on building LLM evaluation systems: LLM-as-judge design, agent and trajectory evals, CI/CD eval gates and benchmark contamination. Every claim is source-linked.

Keploy Java SDK, Initial developer and contributor: Open-source Java SDK for Keploy, an API testing and mocking tool.

Publications

Survey and Evaluation of Food Recommendation, IEEE INDIACom-2016, 2016. ISSN 0973-7529

Certifications

BIAN Banking Architecture Foundation, BIAN Services GmbH, Jul 2025

Education

B.Tech, Computer Science Engineering, Maharaja Agrasen Institute of Technology, New Delhi, 2012 to 2016. Score 80.35%